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Small in-die target design for overlay measurement

Abstract

An overlay metrology system may generate overlay measurements based on isolated images of features on different layers of an overlay target. For example, a target may include a first-layer structure including periodic features with a fine pitch selected to be unresolved by the overlay metrology system, where the gratings structures are oriented to rotate a polarization of rotated diffraction orders relative to a polarization of incident illumination, where the rotated diffraction orders include at least first-order diffraction. In this way, first images generated with a collection polarizer oriented to capture the rotated diffraction orders may include only the first-layer structure, whereas second images generated with a collection polarizer oriented to be aligned with a polarization of incident illumination may include only the second-layer features.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application Ser. No. 63/541,302, filed Sep. 29, 2023, entitled SMALL IN-DIE TARGET DESIGN FOR OVL MEASUREMENT, naming Vladimir Levinski as inventor, which is incorporated herein by reference in the entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to overlay metrology and, more particularly, to an overlay metrology target design utilizing polarization-based isolation of target features when imaged.

BACKGROUND

(2) Many approaches to overlay metrology utilize dedicated overlay targets having features designed for a particular measurement approach. However, a central challenge to the use of dedicated overlay targets relates to variations in printing between device features of interest and the dedicated overlay targets used to measure overlay. This can result in the overlay targets not being fully representative of the device features of interest, which can induce systematic overlay measurement errors.

(3) Efforts to mitigate such errors typically incorporate segmentation of optically-resolvable features with smaller features having a pitch similar to device features of interest. Further efforts to mitigate such errors incorporate overlay targets within semiconductor dies in locations that are relatively close to the device features of interest. Such overlay targets may be referred to as in-die overlay targets or on-chip overlay targets. However, existing in-die overlay target designs still suffer from undesirable overlay measurement errors, which may be associated with coupling between overlay shifts in orthogonal directions (e.g., cross-talk between measurement signals associated with target features on different sample layers) and/or effects of stack variations.

(4) There is therefore a need to develop systems and methods to cure the above deficiencies.

SUMMARY

(5) In embodiments, the techniques described herein relate to an overlay metrology system including a controller including one or more processors configured to execute program instructions causing the one or more processors to receive one or more first images of an overlay target on a sample based on illumination with an illumination beam in accordance with a metrology recipe, where the overlay target includes a first-layer structure on a first layer of the sample, where the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, where the illumination beam is polarized at a 45-degree angle with respect to the first set of periodic features and the second set of periodic features when implementing the metrology recipe, where the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, where the one or more rotated

diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, where the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; where the one or more first images are generated with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders; receive one or more second images of the overlay target based on the illumination with the illumination beam in accordance with the metrology recipe, where the one or more second images are generated with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders; and determine one or more overlay measurements of the sample based on the one or more first images and the one or more second images.

(6) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more first images include the first-layer structure and exclude the second-layer structure.

(7) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more second images include the second-layer structure and exclude the first-layer structure.

(8) In embodiments, the techniques described herein relate to an overlay metrology system, where a size of the overlay target is three micrometers or smaller in any dimension.

(9) In embodiments, the techniques described herein relate to an overlay metrology system, where the first layer of the sample is a process layer, where the second layer of the sample is a resist layer.

(10) In embodiments, the techniques described herein relate to an overlay metrology system, where the first layer of the sample is a resist layer, where the second layer of the sample is a process layer.

(11) In embodiments, the techniques described herein relate to an overlay metrology system, where the first set of periodic features and the second set of periodic features include features with a pitch smaller than a wavelength of the illumination beam.

(12) In embodiments, the techniques described herein relate to an overlay metrology system, where the first-layer structure includes an interleaved pattern of alternating zones of the first set of periodic features and the second set of periodic features, where a pitch of the first set of periodic features and the second set of periodic features is a fine pitch, where the first-layer structure has a first coarse pitch associated with a spatial extent of a first of the zones including at least some of the first set of periodic features and an adjacent one of the zones including at least some of the second set of periodic features.

(13) In embodiments, the techniques described herein relate to an overlay metrology system, where the second-layer structure is non-periodic.

(14) In embodiments, the techniques described herein relate to an overlay metrology system, where the second-layer structure is a cross structure.

(15) In embodiments, the techniques described herein relate to an overlay metrology system, where the illumination beam has a circular profile.

(16) In embodiments, the techniques described herein relate to an overlay metrology system, where the illumination beam has a wedge-shaped profile.

(17) In embodiments, the techniques described herein relate to an overlay metrology system, where an edge of the wedge-shaped profile is oriented parallel to a polarization direction of the illumination beam.

(18) In embodiments, the techniques described herein relate to an overlay metrology system, where an edge of the wedge-shaped profile is oriented orthogonal to a polarization direction of the illumination beam.

(19) In embodiments, the techniques described herein relate to an overlay metrology system including an illumination source configured to generate an illumination beam; an illumination sub-system including one or more lenses to direct the illumination beam to an overlay target on a sample when implementing a metrology recipe, where the overlay target in accordance with the

metrology recipe includes a first-layer structure on a first layer of the sample, where the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, where the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, where the one or more rotated diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, where the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; an imaging sub-system including one or more channels, each including one or more polarizers and one or more detectors, where the one or more channels are configured to generate one or more first images with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders, where the one or more channels are configured to generate one or more second images with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders; and a controller communicatively coupled to the one or more detectors, the controller including one or more processors configured to execute program instructions causing the one or more processors to determine overlay measurements of the sample based on the one or more first images and the one or more second images.

(20) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more first images include the first-layer structure and exclude the second-layer structure.

(21) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more second images include the second-layer structure and exclude the first-layer structure.

(22) In embodiments, the techniques described herein relate to an overlay metrology system, where a size of the overlay target is three micrometers or smaller in any dimension.

(23) In embodiments, the techniques described herein relate to an overlay metrology system, where the first set of periodic features and the second set of periodic features include features with a pitch smaller than a wavelength of the illumination beam.

(24) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more channels include a first channel to generate the one or more first images, where the first channel includes a first of the one or more detectors and a first polarizer aligned with the polarization of the one or more rotated diffraction orders; and a second channel to generate the one or more second images, where the second channel includes a second of the one or more detectors and a second polarizer orthogonal to the polarization of the one or more rotated diffraction orders.

(25) In embodiments, the techniques described herein relate to an overlay metrology system, where the one or more channels includes a single channel with a single polarizer, where the one or more first images are generated with the single polarizer aligned with the polarization of the one or more rotated diffraction orders, where the one or more second images are generated with the single polarizer orthogonal to the polarization of the one or more rotated diffraction orders.

(26) In embodiments, the techniques described herein relate to an overlay metrology system, where the first layer of the sample is a process layer, where the second layer of the sample is a resist layer.

(27) In embodiments, the techniques described herein relate to an overlay metrology system, where the first layer of the sample is a resist layer, where the second layer of the sample is a process layer.

(28) In embodiments, the techniques described herein relate to an overlay metrology system, where the first-layer structure includes an interleaved pattern of alternating zones of the first set of periodic features and the second set of periodic features, where a pitch of the first set of periodic features and the second set of periodic features is a fine pitch, where the first-layer structure has a first coarse pitch associated with a spatial extent of a first of the zones including at least some of the first set of periodic features and an adjacent one of the zones including at least some of the

second set of periodic features.

(29) In embodiments, the techniques described herein relate to an overlay metrology system, where the second-layer structure is non-periodic.

(30) In embodiments, the techniques described herein relate to an overlay metrology system, where the second-layer structure is a cross structure.

(31) In embodiments, the techniques described herein relate to an overlay metrology system, where the illumination beam has a circular profile.

(32) In embodiments, the techniques described herein relate to an overlay metrology system, where the illumination beam has a wedge-shaped profile.

(33) In embodiments, the techniques described herein relate to an overlay metrology system, where an edge of the wedge-shaped profile is oriented parallel to a polarization direction of the illumination beam.

(34) In embodiments, the techniques described herein relate to an overlay metrology system, where an edge of the wedge-shaped profile is oriented orthogonal to a polarization direction of the illumination beam.

(35) In embodiments, the techniques described herein relate to an overlay metrology method including generating one or more first images of an overlay target on a sample based on illumination with an illumination beam in accordance with a metrology recipe, where the overlay target includes a first-layer structure on a first layer of the sample, where the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, where the illumination beam is polarized at a 45-degree angle with respect to the first set of periodic features and the second set of periodic features when implementing the metrology recipe, where the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, where the one or more rotated diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, where the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; where the one or more first images are generated with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders, where the one or more first images include the first-layer structure and exclude the second-layer structure; generating one or more second images of the overlay target based on illumination with the illumination beam in accordance with the metrology recipe, where the one or more second images are generated with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders, where the one or more second images include the second-layer structure and exclude the first-layer structure; and determining overlay measurements of the sample based on the one or more first images and the one or more second images.

(36) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not necessarily restrictive of the invention as claimed. The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and together with the general description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The numerous advantages of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures.

- (2) FIG. 1A illustrates a conceptual view illustrating an overlay metrology system, in accordance with one or more embodiments of the present disclosure.
- (3) FIG. 1B is a conceptual view illustrating the optical sub-system, in accordance with one or more embodiments of the present disclosure.
- (4) FIG. 2A illustrates a top view of a first design of a first-layer structure of an overlay target located on a first layer of the sample, in accordance with one or more embodiments of the present disclosure.
- (5) FIG. 2B illustrates a top view of a second design of first-layer structure of an overlay target located on a first layer of the sample, in accordance with one or more embodiments of the present disclosure.
- (6) FIG. 2C illustrates a top view of a second-layer structure of an overlay target located on a second layer of the sample, in accordance with one or more embodiments of the present disclosure.
- (7) FIG. 3A illustrates an effective medium representation of structures within a coarse pitch of the overlay target in FIG. 2A for light polarized along the X direction, in accordance with one or more embodiments of the present disclosure.
- (8) FIG. 3B illustrates an effective medium representation of structures within a coarse pitch of the overlay target in FIG. 2A for light polarized along the Y direction, in accordance with one or more embodiments of the present disclosure.
- (9) FIG. 4 illustrates a conceptual illustration of an electric field direction of various diffraction orders generated by the first-layer structure illustrated in FIG. 2A upon illumination with an illumination beam having a polarization oriented 45 degrees, in accordance with one or more embodiments of the present disclosure.
- (10) FIG. 5A illustrates an image of the first-layer structure in FIG. 2A generated using an illumination polarizer and a collection polarizer oriented in orthogonal directions, in accordance with one or more embodiments of the present disclosure.
- (11) FIG. 5B illustrates an image of the second-layer structure of FIG. 2C generated with an illumination polarizer and a collection polarizer with parallel orientations, in accordance with one or more embodiments of the present disclosure.
- (12) FIG. 6A illustrates a top view of the overlay target showing the relative positions of the first-layer structure and the second-layer structure, in accordance with one or more embodiments of the present disclosure.
- (13) FIG. 6B illustrates the image of the first-layer structure in FIG. 5A artificially overlapped on the image of the second-layer structure in FIG. 5B, in accordance with one or more embodiments of the present disclosure.
- (14) FIG. 7 is a flow diagram illustrating steps performed in a method for overlay metrology, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(15) Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings. The present disclosure has been particularly shown and described with respect to certain embodiments and specific features thereof. The embodiments set forth herein are taken to be illustrative rather than limiting. It should be readily apparent to those of ordinary skill in the art that various changes and modifications in form and detail may be made without departing from the spirit and scope of the disclosure.

(16) Embodiments of the present disclosure are directed to systems and methods for imaging overlay metrology using overlay targets designed to provide independent imaging of features on different sample layers and further designed to prevent cross-talk between features on different sample layers. It is contemplated herein that cross-talk between features on different sample layers may be increasingly problematic as the overall size of an overlay target is reduced. For example, typical overlay target designs may suffer from unacceptable cross-talk when printed with sizes suitable for in-die placement (e.g., having an overall size of three (3) micrometers per side or less).

However, the systems and methods disclosed herein may enable small overlay targets suitable for in-die placement (e.g., having an overall size of 3 micrometers per side or less) while eliminating cross-talk or at least reducing cross-talk to an acceptable tolerance level. Further, the pitches of periodic features on the target may generally be as small as possible or practicable.

(17) In embodiments, an overlay target may include a first-layer structure on a first layer of the sample, where the first-layer structure includes periodic features designed to selectively rotate a polarization at least some diffraction orders (e.g., at least first-order diffraction) by 90 degrees while maintaining a polarization direction of at least zero-order diffraction. For example, the first-layer structure may rotate at least first-order diffraction (e.g., may rotate first-order diffraction and potentially additional higher-order diffraction) by 90 degrees relative to zero-order diffraction. For convenience, diffraction orders with rotated polarization directions relative to a polarization of incident illumination are referred to as rotated diffraction orders. The overlay target may further include a second-layer structure on a second layer of the sample, where the second-layer structure does not exhibit the polarization-rotating behavior of the first-layer structure.

(18) It is contemplated herein that such a configuration may enable isolated imaging of the first-layer structure and the second-layer structure using polarization-based techniques. For example, an overlay metrology system may illuminate the overlay target with light having a polarization oriented at a 45-degree angle relative to the periodic features of the first-layer structure and one or more polarizers in an imaging sub-system. In particular, adjusting a polarizer in an imaging sub-system at a 135-degree angle relative to the features of the first-layer structure may isolate polarization-rotated first-order diffraction from the first-layer structure such that a resulting image may be associated only with the first-layer structure. Further, adjusting a polarizer in the imaging sub-system to be parallel to the incident illumination may block the polarization-rotated first order light from the first-layer structure but pass light associated with the second-layer structure such that the associated image may be associated only with the second-layer structure. An overlay measurement may then be generated based on centers of symmetry of the overlay target in the two images. Notably, these two isolated images may be generated either sequentially through sequential adjustment of the polarizer in the imaging sub-system or simultaneously using a multi-channel configuration.

(19) Various designs and considerations for overlay targets are generally described in R. Attota, et al., "Evaluation of new in-chip and arrayed line overlay target designs," Proc. SPIE 5375, Metrology, Inspection, and Process Control for Microlithography XVIII, (24 May 2004); Y. S. Ku, "In-chip overlay measurement by existing bright-field imaging optical tools," Proc. SPIE 5752, Metrology, Inspection, and Process Control for Microlithography XIX, (10 May 2005); U.S. Patent Publication No. 2024/0053687 published on Feb. 15, 2024; and U.S. Pat. No. 10,337,991 issued on Jul. 2, 2019; all of which are incorporated herein by reference in their entireties.

(20) Referring now to FIGS. 1A-7, systems and methods providing isolated imaging of features of an overlay target in different sample layers are described in greater detail, in accordance with one or more embodiments of the present disclosure.

(21) FIG. 1A illustrates a conceptual view illustrating an overlay metrology system **100**, in accordance with one or more embodiments of the present disclosure.

(22) In some embodiments, the overlay metrology system **100** includes an optical sub-system **102** to acquire overlay signals from overlay targets based on any number of metrology recipes.

(23) The optical sub-system **102** may include an illumination sub-system **104** to direct at least one illumination beam **106** with a selected polarization to an overlay target **108** on a sample **110**. For example, the illumination sub-system **104** may include an illumination polarizer **112** to ensure that the illumination beam **106** incident on the sample **110** is linearly polarized along a selected direction.

(24) The optical sub-system **102** may further include an imaging sub-system **114** to collect light or other electromagnetic radiation (referred to herein as sample light **116**) emanating from the overlay

target **108**, where at least a portion of the sample light **116** is used to generate one or more images of the overlay target **108** on one or more detectors **118**. Further, the imaging sub-system **114** may include a collection polarizer **120** to control a polarization of light on a detector **118**. For example, as will be described in greater detail herein, isolated images of features of the overlay target **108** on different layers of the sample **110** may be generated with different orientations of the collection polarizer **120**. Further, in some embodiments, the imaging sub-system **114** includes multiple channels with separate collection polarizers **120** to simultaneously capture isolated images of features of the overlay target **108** on different layers of the sample **110** based on different orientations of the respective collection polarizers **120**.

(25) The optical sub-system **102** may be configurable to generate images based on any number of metrology recipes defining measurement parameters for image acquisition and/or design parameters of an overlay target **108** to be imaged. For example, a metrology recipe may include various aspects of an overlay target **108** or a design of an overlay target **108** including, but not limited to, a layout of target features on one or more sample layers, feature sizes, or feature pitches. As another example, a metrology recipe may include illumination parameters such as, but not limited to, an illumination wavelength, an illumination pupil distribution (e.g., a distribution of illumination angles and associated intensities of illumination at those angles), a polarization of incident illumination, a spatial distribution of illumination, or a sample height. By way of another example, a metrology recipe may include collection parameters such as, but not limited to, a collection pupil distribution (e.g., a desired distribution of angular light from the sample **110** to be used for a measurement and associated filtered intensities at those angles), collection field stop settings to select portions of the sample of interest, polarization of collected light, or wavelength filters.

(26) In some embodiments, the overlay metrology system **100** includes a controller **122** communicatively coupled to the optical sub-system **102**. The controller **122** may be configured to direct the optical sub-system **102** to generate overlay signals based on one or more selected metrology recipes. The controller **122** may be further configured to receive data including, but not limited to, images of the overlay target **108** (or structures on different layers thereof) from a detector **118**. Additionally, the controller **122** may be configured to determine overlay associated with an overlay target **108** based on the acquired overlay signals.

(27) In some embodiments, the controller **122** includes one or more processors **124**. For example, the one or more processors **124** may be configured to execute a set of program instructions maintained in a memory **126**, or memory device. The one or more processors **124** of a controller **122** may include any processing element known in the art. In this sense, the one or more processors **124** may include any microprocessor-type device configured to execute algorithms and/or instructions. Further, the memory **126** may include any storage medium known in the art suitable for storing program instructions executable by the associated one or more processors **124**. For example, the memory **126** may include a non-transitory memory medium. As an additional example, the memory **126** may include, but is not limited to, a read-only memory, a random-access memory, a magnetic or optical memory device (e.g., disk), a magnetic tape, a solid-state drive and the like. It is further noted that memory **126** may be housed in a common controller housing with the one or more processors **124**.

(28) Referring now to FIGS. 2A-6B, an overlay target **108** is described, in accordance with one or more embodiments of the present disclosure.

(29) FIG. 2A illustrates a top view of a first design of a first-layer structure **202** of an overlay target **108** located on a first layer of the sample **110**, in accordance with one or more embodiments of the present disclosure. FIG. 2B illustrates a top view of a second design of first-layer structure **202** of an overlay target **108** located on a first layer of the sample **110**, in accordance with one or more embodiments of the present disclosure. FIG. 2C illustrates a top view of a second-layer structure **204** of an overlay target **108** located on a second layer of the sample **110**, in accordance with one or

more embodiments of the present disclosure.

(30) In embodiments, the first-layer structure **202** includes periodic features characterized by a fine pitch **206** (e.g., fine period, a segmentation pitch, a segmentation period, or the like) and coarse pitch **208** (e.g., a coarse period). For example, the first-layer structure **202** may include a first set of periodic features **210** and a second set of periodic features **212**, where a direction of periodicity of the first set of periodic features **210** is orthogonal to (e.g., at a 90-degree angle to) a direction of periodicity of the second set of periodic features **212**. Further, features within a coarse pitch **208** may be further arranged into two zones **214** (e.g., two adjacent zones **214** may form the coarse pitch **208**), each having different polarization directions. Put another way, the coarse pitch **208** may correspond to a spatial extent of two adjacent zones **214**.

(31) For example, box **216** in FIGS. 2A and 2B depicts features within a period associated with the coarse pitch **208**, which are divided into a first zone **214a** with some of the first set of periodic features **210** and a second zone **214b** with some of the second set of periodic features **212**. Further, the first-layer structure **202** may include an interleaved pattern of alternating zones **214**. For example, FIGS. 2A and 2B depict configurations in which zones **214a,b** including features with orthogonal periodicity are distributed along different diagonals of the overlay target **108**. In this configuration, a coarse pitch **208** along both the X and Y directions includes two zones **214a,b** having orthogonal periodicity with the fine pitch **206**.

(32) In embodiments, the fine pitch **206** is selected to be smaller than a wavelength of the illumination beam **106** of the optical sub-system **102** and in some cases substantially smaller than the wavelength of the illumination beam **106**. Put another way, the fine pitch **206** may be selected to be unresolvable by the imaging sub-system **114** of the overlay metrology system **100**. However, this is not a requirement. In some embodiments, the fine pitch **206** is resolvable by the imaging sub-system **114**.

(33) In embodiments, any of the zones **214a,b** may be large enough to be resolvable by the imaging sub-system **114**. In this way, although the periodic features within the zones having the fine pitch **206** may not be resolvable, the zones **214a,b** themselves may be resolvable and thus provide a measurable signal in an image of the overlay target **108**. However, in some embodiments, the first-layer structure **202** (and the overlay target **108** as a whole) is printed with dimensions suitable for placement within die regions of the sample **110** (e.g., close to device features). For example, the first-layer structure **202** (and the overlay target **108** as a whole) may have a size along any or all dimensions of 3 micrometers or less.

(34) It is contemplated herein that, when the illumination beam **106** has a particular polarization, the first-layer structure **202** may generate diffraction orders associated with the fine pitch **206**, where a polarization direction of at least some non-zero diffraction orders are rotated with respect to zero-order diffraction. For example, the first set of periodic features **210** and the second set of periodic features **212** may rotate a polarization direction of at least first-order diffraction from the first-layer structure **202** relative to zero-order diffraction from the first-layer structure **202**. As described previously herein, the diffraction orders with rotated polarization (e.g., at least the first-order diffraction from the first-layer structure **202**) are referred to as rotated diffraction orders.

(35) For example, the first-layer structure **202** in FIG. 2A may rotate a polarization of all diffraction except zero-order diffraction by 90 degrees when the polarization of the illumination beam **106** is oriented at 45 degrees or 135 degrees (e.g., at a 45-degree angle with respect to the directions of periodicity of the first set of periodic features **210** and the second set of periodic features **212** in this design). As another example, the first-layer structure **202** in FIG. 2B may rotate a polarization of odd diffraction orders (e.g., first-order diffraction, third-order diffraction, and the like) by 90 degrees when the polarization of the illumination beam **106** is oriented at 0 degrees or 90 degrees (e.g., first-order diffraction, third-order diffraction, and the like), but zero-order diffraction as well as even-order diffraction (e.g., second-order diffraction, fourth-order diffraction, or the like) may have the same polarization direction as the incident illumination beam **106**.

(36) The polarization-rotating properties of the first-layer structure **202** may be understood by considering the effective permittivities of the different zones of periodic features. FIGS. **3A-4** depict the polarization behavior and polarization-dependent imaging of an overlay target **108** with first-layer structure **202** arranged according to FIG. **2A**.

(37) FIG. **3A** illustrates an effective medium representation of structures within a coarse pitch **208** of the overlay target **108** in FIG. **2A** (e.g., features within the box **216**) for light polarized along the X direction, in accordance with one or more embodiments of the present disclosure. FIG. **3B** illustrates an effective medium representation of structures within a coarse pitch **208** of the overlay target **108** in FIG. **2A** (e.g., features within the box **216**) for light polarized along the Y direction, in accordance with one or more embodiments of the present disclosure.

(38) It is contemplated herein that the first-layer structure **202** operates as a half-wave plate for all diffraction orders apart from zero order diffraction.

(39) In FIGS. **3A** and **3B**, the effective medium for permittivity are defined by the following formulae:

$$(40) \quad \epsilon_{\text{sub}} = \epsilon_{\text{Gr}} \cdot \text{Math.} + \epsilon_{\text{sur}} \cdot \text{Math.} (1 - \eta) \quad (1) \quad \epsilon_{\text{sub}} = \frac{\epsilon_{\text{Gr}} \cdot \text{Math.}}{\epsilon_{\text{Gr}} \cdot \text{Math.} (1 - \eta) + \epsilon_{\text{sur}} \cdot \text{Math.}} \cdot (2)$$

where $\epsilon_{\text{sub}}.$ Gr is a permittivity of a material forming the grating features (e.g., the first set of periodic features **210** and the second set of periodic features **212**), $\epsilon_{\text{sub}}.$ sur is a permittivity of material surrounding the grating features, and η is a fraction of the coarse pitch **208** associated with the segmentation with the fine pitch **206** along the X direction (e.g., a width of the first zone **214a** in FIG. **2A** relative to the coarse pitch **208**). Put another way, η corresponds to a duty cycle of the orthogonal fine segmentation.

(41) As illustrated by equations (1) and (2) along with FIGS. **3A-3B**, arranging features on a layer of an overlay target **108** to provide both a coarse pitch **208** and zones of orthogonal fine pitch **206** effectively operates as a wave plate and provides different phase shifts for light polarized along the orthogonal directions.

(42) FIG. **4** illustrates a conceptual illustration of an electric field direction (e.g., polarization direction) of various diffraction orders generated by the first-layer structure **202** illustrated in FIG. **2A** upon illumination with an illumination beam **106** having a polarization oriented 45 degrees (e.g., with respect to the first set of periodic features **210** and the second set of periodic features **212**), in accordance with one or more embodiments of the present disclosure. As illustrated in FIG. **4**, zero-order diffraction **402** may have the same electric field direction as the incident illumination beam **106**, whereas non-zero diffraction orders **404** may be orthogonally polarized to the zero-order diffraction **402**.

(43) As a result, a collection polarizer **120** in the imaging sub-system **114** oriented at 135 degrees may block the zero-order diffraction **402** and isolate the rotated non-zero diffraction orders **404** without reducing their amplitudes.

(44) FIG. **5A** illustrates an image **502** of the first-layer structure **202** in FIG. **2A** generated using an illumination polarizer **112** and a collection polarizer **120** oriented in orthogonal directions, in accordance with one or more embodiments of the present disclosure. For example, the image **502** may be generated using an illumination polarizer **112** oriented at 45 degrees and a collection polarizer **120** oriented at 135 degrees. Using this cross-polarization scheme, a detector **118** may register a whole signal associated with the polarization-rotated diffraction orders without zero-order diffraction **402** (which may be referred to as DC components or constant components).

(45) Further, the contrast of the image **502** may depend on the materials used to fabricate the first-layer structure **202**. In a general sense, the first-layer structure **202** may be fabricated in any sample layer including a process layer or a resist layer. However, contrast may be improved in a process layer. As an example, if the first-layer structure **202** is printed with silicon features surrounded by an oxide, $\epsilon_{\text{sub}}.$ Gr=3.8 and $\epsilon_{\text{sub}}.$ sur=1.45. For a 1:1 duty cycle of the zones **214a,b** within a coarse pitch **208**, Equations (1) and (2) provide that an effective permittivity of the first zone **214a** with

some of the first set of periodic features **210** is $\epsilon_{\text{sub.1}}=2.63$, whereas an effective permittivity of the second zone **214b** with some of the second set of periodic features **212** is $\epsilon_{\text{sub.2}}=2.1$. This difference in effective permittivities provides similar contrast in the image **502** as typical non-segmented targets in a resist layer and may thus be easily observable.

(46) Referring now to FIGS. **2C** and **5B**, imaging of second-layer structure **204** on a second layer of the sample **110** is described in greater detail, in accordance with one or more embodiments of the present disclosure.

(47) In embodiments, the second-layer structure **204** may include any distribution of features that maintains a polarization of an incident illumination beam **106**. Put another way, the second-layer structure **204** may include any distribution of features that does not rotate the polarization of some diffraction orders in a manner similar to the first-layer structure **202**. For instance, the second-layer structure **204** may incorporate features in any standard periodic or non-periodic target design that maintains a polarization of an incident illumination beam **106** (e.g., provides diffraction with an unmodified polarization direction).

(48) FIG. **2C** depicts a non-limiting configuration in which the second-layer structure **204** is a cross structure centered on the overlay target **108**.

(49) As a result, the first-layer structure **202** and the second-layer structure **204** may be independently imaged using different polarizations. An overlay measurement may then be generated using typical techniques based on the two images generated at the orthogonal polarizations. For example, an overlay measurement may be generated based on a difference between a center of symmetry of the first-layer structure **202** from a first image and a center of symmetry of the second-layer structure **204** from a second image. It is contemplated herein that mitigation of such cross-talk may increase the accuracy of an overlay measurement generated based on these two images relative to alternative techniques that suffer from cross-talk between light interacting with features on different layers of a target.

(50) For example, the image **502** of the first-layer structure **202** in FIG. **5A** generated with an illumination polarizer **112** oriented at 45 degrees and a collection polarizer **120** oriented at 135 degrees may not include the second-layer structure **204** since the light interacting with this second-layer structure **204** maintains the polarization state of the incident illumination beam **106** and is blocked by the collection polarizer **120**.

(51) In contrast, an image generated with the collection polarizer **120** oriented at 45 degrees (e.g., parallel to the illumination polarizer **112**) may include only the second-layer structure **204** and not the first-layer structure **202**. Rather, the light passed by the collection polarizer **120** in this orientation include light interacting with the second-layer structure **204** as well as zero-order diffraction. Although a portion of this zero-order diffraction may be associated with the first-layer structure **202** (e.g., the zero-order diffraction **402** depicted in FIG. **4**), this light is insufficient to resolve any portion of the first-layer structure **202**.

(52) FIG. **5B** illustrates an image **504** of the second-layer structure **204** of FIG. **2C** generated with an illumination polarizer **112** and a collection polarizer **120** with parallel orientations, in accordance with one or more embodiments of the present disclosure. For example, the image **504** may be generated with the illumination polarizer **112** and the collection polarizer **120** both oriented at 45 degrees.

(53) Referring now to FIGS. **6A** and **6B**, it is contemplated herein that cross-talk between the first-layer structure **202** and the second-layer structure **204** may be further mitigated by reducing the physical overlap between these structures. In this way, the amount of light that interacts with both structures (e.g., double diffraction) may be limited.

(54) The second-layer structure **204** may further be configured to provide that features in the second-layer structure **204** have relatively little overlap with features of the first-layer structure **202** (e.g., the first set of periodic features **210** and the second set of periodic features **212**). For example, a design of the overlay target **108** (e.g., in accordance with a metrology recipe) may be selected to

provide that an area of the overlay target **108** in which the features of the second-layer structure **204** overlap with features of the first-layer structure **202** may be below a selected threshold (e.g., 20%, 15%, 10%, 5%, 1%, or any suitable value).

(55) FIG. **6A** illustrates a top view of the overlay target **108** showing the relative positions of the first-layer structure **202** and the second-layer structure **204**, in accordance with one or more embodiments of the present disclosure. As shown, this design provides that the second-layer structure **204** lies substantially within spaces between the zones **214** or edges of the zones **214** of the first-layer structure **202**.

(56) FIG. **6B** illustrates the image **502** of the first-layer structure **202** in FIG. **5A** artificially overlapped on the image **504** of the second-layer structure **204** in FIG. **5B**, in accordance with one or more embodiments of the present disclosure. As described previously herein, the second-layer structure **204** is not actually visible on the image **502**. Rather, FIG. **6B** illustrates how the presence of the second-layer structure **204** in this particular design of the second-layer structure **204** may provide relatively little impact on the image **502** of the first-layer structure **202**.

(57) As shown in FIGS. **6A-6B**, the particular configuration of a cross-shaped second-layer structure **204** provides relatively little overlap between the second-layer structure **204** and the first-layer structure **202**, which reduces an amount of light that interacts with both structures (e.g., double diffraction) and thus effectively mitigates cross-talk. However, this design of the second-layer structure **204** is merely an illustration and the second-layer structure **204** may have any design providing suppression of cross-talk between images taken with orthogonal polarizations.

(58) Referring generally to FIGS. **3A-6B**, it is to be understood that FIGS. **3A-6B** are provided solely for illustrative purposes and should not be interpreted as limiting the scope of the present disclosure. For example, FIGS. **3A-6B** illustrate the particular non-limiting configuration of an overlay target **108** including a first-layer structure **202** arranged according to FIG. **2A**. However, any overlay target **108** with any design of a first-layer structure **202** that provides polarization rotation of select diffraction orders is within the spirit and scope of the present disclosure. Further, the principles described with respect to FIGS. **3A-6B** may be extended to any such design of an overlay target **108**.

(59) For example, it is contemplated herein that some designs of an overlay target **108** may not fully eliminate cross-talk, but may rather suppress cross-talk for at least one of the images. For instance, the polarization-rotating properties of the first-layer structure **202** may generally provide that the first images generated with cross-polarizations of the illumination beam **106** and the sample light **116** display effective suppression of signals associated with the second-layer structure **204**. However, there may be some cross-talk associated with the second-layer structure **204** in the first images. In some embodiments, a design of the overlay target **108** (e.g., in accordance with a metrology recipe) may be selected to suppress signals associated with the second-layer structure **204** in the first images below a selected threshold. Similarly, the second images generated with parallel polarizers of the illumination beam **106** and the sample light **116** display effective suppression of signals associated with the first-layer structure **202**. However, there may be some cross-talk associated with the first-layer structure **202** in the second images. In some embodiments, a design of the overlay target **108** (e.g., in accordance with a metrology recipe) may be selected to suppress signals associated with the first-layer structure **202** in the second images below a selected threshold. In either or both cases, suppression of cross-talk may be characterized using any suitable metric. For example, cross-talk may be characterized by a metric describing a strength of a signal associated with undesired features in an image (e.g., a strength of a signal associated with the second-layer structure **204** in a first image and/or a strength of a signal associated with the first-layer structure **202** in a second image). As another example, cross-talk may be characterized by a metric describing a contrast of desired features in an image (e.g., a contrast of first-layer structure **202** in a first image and/or a contrast of second-layer structure **204** in a second image). Regardless of the metric, it is contemplated herein that systems and methods disclosed herein may reduce

cross-talk by an order of magnitude relative to existing target designs in some embodiments.

(60) Referring now to FIG. 1B, additional aspects of the overlay metrology system **100** are described in greater detail, in accordance with one or more embodiments of the present disclosure.

(61) FIG. 1B is a conceptual view illustrating the optical sub-system **102**, in accordance with one or more embodiments of the present disclosure. In some embodiments, the optical sub-system **102** includes an illumination source **128** configured to generate an illumination beam **106**. The illumination beam **106** may include one or more selected wavelengths of light including, but not limited to, ultraviolet (UV) radiation, visible radiation, or infrared (IR) radiation.

(62) The illumination source **128** may include any type of illumination source suitable for providing an illumination beam **106**. In some embodiments, the illumination source **128** is a laser source. For example, the illumination source **128** may include, but is not limited to, one or more narrowband laser sources, a broadband laser source, a supercontinuum laser source, a white light laser source, or the like. In this regard, the illumination source **128** may provide an illumination beam **106** having high coherence (e.g., high spatial coherence and/or temporal coherence). In some embodiments, the illumination source **128** includes a laser-sustained plasma (LSP) source. For example, the illumination source **128** may include, but is not limited to, a LSP lamp, a LSP bulb, or a LSP chamber suitable for containing one or more elements that, when excited by a laser source into a plasma state, may emit broadband illumination. In some embodiments, the illumination source **128** includes a lamp source. For example, the illumination source **128** may include, but is not limited to, an arc lamp, a discharge lamp, an electrode-less lamp, or the like. In this regard, the illumination source **128** may provide an illumination beam **106** having low coherence (e.g., low spatial coherence and/or temporal coherence).

(63) In some embodiments, the illumination sub-system **104** directs the illumination beam **106** to the sample **110** or a portion thereof including at least one overlay target **108**. The illumination sub-system **104** may include one or more optical components suitable for modifying and/or conditioning the illumination beam **106** as well as directing the illumination beam **106** to the sample **110**. For example, the illumination sub-system **104** may include, but is not required to include, one or more lenses **130** (e.g., to collimate the illumination beam **106**, to relay pupil and/or field planes, or the like), an illumination polarizer **112** to adjust the polarization of the illumination beam **106**, one or more filters, one or more beam splitters, one or more diffusers, one or more homogenizers, one or more apodizers, one or more beam shapers, or one or more mirrors (e.g., static mirrors, translatable mirrors, scanning mirrors, or the like). In some embodiments, the optical sub-system **102** includes an objective lens **132** to focus the illumination beam **106** onto the sample **110** (e.g., onto the overlay target **108**).

(64) The illumination beam **106** may have any spatial profile. For example, the spatial profile may be circular. As another example, the spatial profile may be structured with a wedge profile or any other suitable shape. It is contemplated herein that structured illumination with a wedge-shaped profile with an edge either parallel to or perpendicular to a polarization direction may further suppress cross-talk in a cross-polarization imaging configuration as disclosed throughout the present disclosure.

(65) The shape of the illumination beam **106** on the sample **110** may be controlled using any suitable optical elements within the illumination sub-system **104** such as, but not limited to, one or more stops in an illumination field plane conjugate to the sample **110**.

(66) The illumination beam **106** may have any number of poles (e.g., established in an illumination pupil plane) providing illumination at any number or range of altitude or azimuth incidence angles. In some embodiments, the illumination beam **106** may include one or more poles located in an illumination pupil at that coincide with an electric field direction (e.g., a polarization direction) of the associated light.

(67) In some embodiments, the sample **110** is disposed on a sample stage **134** suitable for securing the sample **110** and further configured to position the sample **110** with respect to the illumination

beam **106**.

(68) In some embodiments, the imaging sub-system **114** includes one or more detectors **118** configured to capture sample light **116** emanating from the sample **110** and generate one or more images of the overlay target **108**, or portions thereof. The imaging sub-system **114** may include multiple optical elements to direct and/or modify illumination collected by the objective lens **132** including, but not limited to one or more lenses **136**, one or more filters, one or more collection polarizers **120**, one or more beam blocks, or one or more beamsplitters.

(69) The illumination sub-system **104** and the imaging sub-system **114** of the optical sub-system **102** may be oriented in a wide range of configurations suitable for illuminating the overlay target **108** with the illumination beam **106** and collecting the sample light **116** in response to the incident illumination beam **106**. For example, as illustrated in FIG. 1B, the optical sub-system **102** may include a beamsplitter **138** oriented such that the objective lens **132** may simultaneously direct the illumination beam **106** to the overlay target **108** and collect radiation emanating from the sample **110**. By way of another example, the illumination sub-system **104** and the imaging sub-system **114** may contain non-overlapping optical paths.

(70) As described throughout the present disclosure, isolated images of features of an overlay target **108** located on different layers of the sample **110** may be generated using specific orientations of an illumination polarizer **112** in the illumination sub-system **104** and a collection polarizer **120** in the imaging sub-system **114**. For example, isolated images of features of an overlay target **108**, including the first-layer structure **202** as illustrated in FIG. 2A, may be provided with an illumination polarizer **112** oriented at 45 degrees and a collection polarizer **120** oriented at 135 degrees. As another example, isolated images of features of an overlay target **108** including first-layer structure **202** as illustrated in FIG. 2B may be provided with an illumination polarizer **112** oriented at 0 degrees and a collection polarizer **120** oriented at 90 degrees.

(71) In some embodiments, two isolated images are generated sequentially on a single detector **118** (e.g., in a single channel) by sequentially adjusting the orientation of a single collection polarizer **120**. In some embodiments, two isolated images are generated simultaneously on two detectors **118** in different channels of the imaging sub-system **114**, where the two channels include collection polarizers **120** oriented to capture the two isolated images.

(72) As an illustration, FIG. 1B depicts a configuration in which the imaging sub-system **114** includes a beamsplitter **140** (e.g., a non-polarizing beamsplitter) to split the sample light **116** into a first channel **142a** and a second channel **142b**. In this way, the collection polarizers **120** in the different channels **142a,b** may have orthogonal polarizations (e.g., oriented at a 90-degree with respect to each other) oriented to generate isolated images of the first-layer structure **202** and the second-layer structure **204** of an overlay target **108**.

(73) FIG. 7 is a flow diagram illustrating steps performed in a method **700** for overlay metrology, in accordance with one or more embodiments of the present disclosure. The embodiments and enabling technologies described previously herein in the context of the overlay metrology system **100** should be interpreted to extend to the method **700**. For example, various components of the overlay metrology system **100** may implement one or more steps of the method **700** individually or in combination. In some embodiments, the processors **124** of the controller **122** may execute program instructions causing the processors **124** to implement one or more steps of the method **700** either directly or indirectly by providing control signals to one or more additional components of the overlay metrology system **100** to implement one or more steps of the method **700**. It is further noted, however, that the method **700** is not limited to the architecture of the overlay metrology system **100**.

(74) In embodiments, the method **700** includes a step **702** of generating a first image of an overlay target **108** on a sample **110** with a first-layer structure **202** including polarization-rotating features and a second-layer structure **204** including polarization-maintaining features, where the first image is generated with polarization-filtered light aligned with a polarization direction of the first-order

diffraction from a first-layer structure **202** (e.g., aligned with a polarization of rotated diffraction orders more generally). In this way, the one or more first images include the first-layer structure and exclude the second-layer structure.

(75) In embodiments, the method **700** includes a step **704** of generating a second image of the overlay target **108** based on illumination with the illumination beam **106**, where the second image is generated with polarization-filtered light aligned orthogonal to the polarization direction of the first-order diffraction from the first-layer structure **202** (e.g., orthogonal to a polarization of the rotated diffraction orders more generally). In this way, the one or more second images include the second-layer structure and exclude the first-layer structure.

(76) The first and second images may be generated simultaneously or sequentially with one or more collection polarizers **120**.

(77) In embodiments, the method **700** includes a step **706** of determining overlay measurements of the sample based on the first image and the second image. Any suitable technique may be used such as, but not limited to, determining overlay based on a difference between a center of symmetry of the first-layer structure **202** in the first image and a center of symmetry of the second-layer structure **204**

(78) In some embodiments, the method **700** further includes a step of generating one or more correctables to control one or more processing tools, such as, but not limited to a lithography tool (e.g., a scanner, a stepper, or the like), an etching tool, a polishing tool, or the like. For example, correctables may be generated based on overlay measurements associated with one or more overlay targets distributed across the sample. Further, the correctables may be associated with any combination of feedback or feed-forward control. As an illustration, feedback control may correct for deviations of a process tool between different samples in a particular lot or sequence of lots. As another illustration, feed-forward control may provide adjustments for a subsequent process step based on measurements of a current process step.

(79) The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected” or “coupled” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “couplable” to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to physically interactable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interactable and/or logically interacting components.

(80) It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction, and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes. Furthermore, it is to be understood that the invention is defined by the appended claims.

Claims

1. An overlay metrology system comprising: a controller including one or more processors configured to execute program instructions causing the one or more processors to: receive one or

more first images of an overlay target on a sample based on illumination with an illumination beam in accordance with a metrology recipe, wherein the overlay target comprises: a first-layer structure on a first layer of the sample, wherein the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, wherein the illumination beam is polarized at a 45-degree angle with respect to the first set of periodic features and the second set of periodic features when implementing the metrology recipe, wherein the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, wherein the one or more rotated diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, wherein the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; wherein the one or more first images are generated with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders; receive one or more second images of the overlay target based on the illumination with the illumination beam in accordance with the metrology recipe, wherein the one or more second images are generated with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders; and determine one or more overlay measurements of the sample based on the one or more first images and the one or more second images.

2. The overlay metrology system of claim 1, wherein a size of the overlay target is three micrometers or smaller in any dimension.
3. The overlay metrology system of claim 1, wherein the first layer of the sample is a process layer, wherein the second layer of the sample is a resist layer.
4. The overlay metrology system of claim 1, wherein the first layer of the sample is a resist layer, wherein the second layer of the sample is a process layer.
5. The overlay metrology system of claim 1, wherein the first set of periodic features and the second set of periodic features include features with a pitch smaller than a wavelength of the illumination beam.
6. The overlay metrology system of claim 1, wherein the illumination beam has a circular profile.
7. The overlay metrology system of claim 1, wherein the illumination beam has a wedge-shaped profile.
8. The overlay metrology system of claim 7, wherein an edge of the wedge-shaped profile is oriented parallel to a polarization direction of the illumination beam.
9. The overlay metrology system of claim 7, wherein an edge of the wedge-shaped profile is oriented orthogonal to a polarization direction of the illumination beam.
10. The overlay metrology system of claim 1, wherein the one or more first images include the first-layer structure and exclude the second-layer structure.
11. The overlay metrology system of claim 10, wherein the one or more second images include the second-layer structure and exclude the first-layer structure.
12. The overlay metrology system of claim 1, wherein the first-layer structure includes an interleaved pattern of alternating zones of the first set of periodic features and the second set of periodic features, wherein a pitch of the first set of periodic features and the second set of periodic features is a fine pitch, wherein the first-layer structure has a first coarse pitch associated with a spatial extent of a first of the zones including at least some of the first set of periodic features and an adjacent one of the zones including at least some of the second set of periodic features.
13. The overlay metrology system of claim 12, wherein the second-layer structure is non-periodic.
14. The overlay metrology system of claim 13, wherein the second-layer structure is a cross structure.
15. An overlay metrology system comprising: an illumination source configured to generate an

illumination beam; an illumination sub-system including one or more lenses to direct the illumination beam to an overlay target on a sample when implementing a metrology recipe, wherein the overlay target in accordance with the metrology recipe comprises: a first-layer structure on a first layer of the sample, wherein the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, wherein the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, wherein the one or more rotated diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, wherein the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; an imaging sub-system including one or more channels, each including one or more polarizers and one or more detectors, wherein the one or more channels are configured to generate one or more first images with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders, wherein the one or more channels are configured to generate one or more second images with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders; and a controller communicatively coupled to the one or more detectors, the controller including one or more processors configured to execute program instructions causing the one or more processors to determine overlay measurements of the sample based on the one or more first images and the one or more second images.

16. The overlay metrology system of claim 15, wherein a size of the overlay target is three micrometers or smaller in any dimension.

17. The overlay metrology system of claim 15, wherein the first set of periodic features and the second set of periodic features include features with a pitch smaller than a wavelength of the illumination beam.

18. The overlay metrology system of claim 15, wherein the one or more channels comprise: a first channel to generate the one or more first images, wherein the first channel includes a first of the one or more detectors and a first polarizer aligned with the polarization of the one or more rotated diffraction orders; and a second channel to generate the one or more second images, wherein the second channel includes a second of the one or more detectors and a second polarizer orthogonal to the polarization of the one or more rotated diffraction orders.

19. The overlay metrology system of claim 15, wherein the one or more channels includes a single channel with a single polarizer, wherein the one or more first images are generated with the single polarizer aligned with the polarization of the one or more rotated diffraction orders, wherein the one or more second images are generated with the single polarizer orthogonal to the polarization of the one or more rotated diffraction orders.

20. The overlay metrology system of claim 15, wherein the first layer of the sample is a process layer, wherein the second layer of the sample is a resist layer.

21. The overlay metrology system of claim 15, wherein the first layer of the sample is a resist layer, wherein the second layer of the sample is a process layer.

22. The overlay metrology system of claim 15, wherein the illumination beam has a circular profile.

23. The overlay metrology system of claim 15, wherein the illumination beam has a wedge-shaped profile.

24. The overlay metrology system of claim 23, wherein an edge of the wedge-shaped profile is oriented parallel to a polarization direction of the illumination beam.

25. The overlay metrology system of claim 23, wherein an edge of the wedge-shaped profile is oriented orthogonal to a polarization direction of the illumination beam.

26. The overlay metrology system of claim 15, wherein the one or more first images include the first-layer structure and exclude the second-layer structure.
27. The overlay metrology system of claim 26, wherein the one or more second images include the second-layer structure and exclude the first-layer structure.
28. The overlay metrology system of claim 15, wherein the first-layer structure includes an interleaved pattern of alternating zones of the first set of periodic features and the second set of periodic features, wherein a pitch of the first set of periodic features and the second set of periodic features is a fine pitch, wherein the first-layer structure has a first coarse pitch associated with a spatial extent of a first of the zones including at least some of the first set of periodic features and an adjacent one of the zones including at least some of the second set of periodic features.
29. The overlay metrology system of claim 28, wherein the second-layer structure is non-periodic.
30. The overlay metrology system of claim 29, wherein the second-layer structure is a cross structure.
31. An overlay metrology method comprising: generating one or more first images of an overlay target on a sample based on illumination with an illumination beam in accordance with a metrology recipe, wherein the overlay target comprises: a first-layer structure on a first layer of the sample, wherein the first-layer structure includes a first set of periodic features and a second set of periodic features oriented at a 90-degree angle with respect to the first set of periodic features, wherein the illumination beam is polarized at a 45-degree angle with respect to the first set of periodic features and the second set of periodic features when implementing the metrology recipe, wherein the first set of periodic features and the second set of periodic features are arranged in accordance with the metrology recipe to rotate a polarization of one or more rotated diffraction orders from the first-layer structure relative to a polarization of zero-order diffraction, wherein the one or more rotated diffraction orders include at least first-order diffraction from the first-layer structure; and a second-layer structure on a second layer of the sample, wherein the second-layer structure includes one or more features providing an unmodified polarization direction for light emanating from the second-layer structure in response to the illumination beam; wherein the one or more first images are generated with first polarization-filtered light aligned with the polarization of the one or more rotated diffraction orders, wherein the one or more first images include the first-layer structure and exclude the second-layer structure; generating one or more second images of the overlay target based on the illumination with the illumination beam in accordance with the metrology recipe, wherein the one or more second images are generated with second polarization-filtered light aligned orthogonal to the polarization of the one or more rotated diffraction orders, wherein the one or more second images include the second-layer structure and exclude the first-layer structure; and determining overlay measurements of the sample based on the one or more first images and the one or more second images.
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